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THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test – Bhaskara Contest

NMTC at JUNIOR LEVEL – IX & X Standards

Saturday, 27th August, 2016

Note:

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
2. Diagrams are only visual aids; they are NOT drawn to scale.
3. You are free to do rough work on separate sheets.
4. Duration of the test: 2 pm to 4 pm – 2 hours.

PART – A

Note

- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
- For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{2}$ mark.**

1. The sum of the values x, y that satisfy the equations

$$(x+y)2^{y-x} = 1, \quad (x+y)^{x-y} = 2$$

simultaneously is

A. 2 B. $\frac{3}{2}$ C. $\frac{5}{2}$ D. $\frac{7}{2}$

2. If

$$\left(2 - \frac{a}{4} - \frac{4}{a}\right)\left\{(a-4)\sqrt[3]{(a-4)^{-3}} - \frac{(a^2-16)^{-\frac{1}{2}}(a-4)^{-\frac{1}{2}}}{(a+4)^{-\frac{3}{2}}}\right\}\left(\frac{a+4}{a-4}\right) = 2016$$

the value of a is

A. $\frac{4}{1007}$ B. $\frac{3}{2016}$ C. $\frac{4}{2017}$ D. none of these

3. $ABCD$ is a square inscribed in a circle of radius 1 unit. The tangent to the circle at C meets AB produced at P. The length of PD is

A. 2 B. 3 C. $\sqrt{13}$ D. $\sqrt{10}$

4. Quadrilateral $ABCD$ is inscribed in a circle with radius 1 unit. AC is the diameter of the circle and $BD = AB$. The diagonals cut at P. If $PC = \frac{2}{5}$ then the length of CD is equal to

A. $\frac{2}{3}$ B. $\frac{2}{7}$ C. $\frac{1}{8}$ D. $\frac{3}{4}$

5. The number of natural numbers n for which the expression $\dfrac{23n^2+18n+4}{n}$ is also a natural number is

A. 3 B. 2 C. 1 D. 0

6. The cost price of 16 oranges is equal to the selling price of 12 oranges. Then there is a

A. 40% profit B. 20% loss C. $33\frac{1}{3}$ % profit D. $23\frac{1}{3}$ % profit

7. The number of positive integer pairs (a, b) such that $ab - 24 = 2b$ is

A. 6 B. 7 C. 8 D. 9

8. $A = (2+1)(2^2+1)(2^4+1)\cdots(2^{2016}+1)$. The value of $(A+1)^{1/2016}$ is

A. 4 B. 2016 C. 2^{4032} D. 2

9. The sum of two numbers a, b where $a < b$ is 1215 and their H.C.F is 81. The number of pairs of such pairs (a, b) is

A. 1 B. 2 C. 3 D. 4

10. The first Republic Day of India was celebrated on 26th January 1950. What was the day of the week on that date?

A. Tuesday B. Wednesday C. Thursday D. Friday

11. The 12 numbers $a_1, a_2, \ldots, a_{12}$ are in arithmetical progression. The sum of all these numbers is 354. Let $P = a_2 + a_4 + \cdots + a_{12}$ and $Q = a_1 + a_3 + \cdots + a_{11}$. If the ratio $P : Q$ is $32 : 27$, the common difference of the progression is

A. 2 B. 3 C. 4 D. 5

12. A shopkeeper marks the prices of his goods at 20% higher than the original price. There is an increase in demand of the goods, and he further increases the price by 20%. The total profit % is

A. 40 B. 38 C. 42 D. 44

13. A circle passes through the vertices A and D and touches the side BC of a square. The side of the square is 2 cm. The radius of the circle (in cm) is

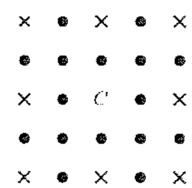
A. $\frac{5}{4}$ B. $\frac{4}{5}$ C. 1 D. $\frac{5}{2}$

14. There are four balls – one green, one red, one blue and one yellow and there are four boxes – one green, one red, one blue and one yellow. A child playing with the balls decides to put the balls in the boxes, one ball in each box. The number of ways in which the child can put the balls in the boxes such that no ball is in a box of its own color is

A. 12 B. 9 C. 24 D. 6

15. The 5×5 array of dots represents trees in an orchard. If you were standing at the central spot marked C, you would not be able to see 8 of the 24 trees (shown as X). If you were standing at the center of a 9×9 array of trees, how many of the 80 trees would be hidden?

A. 40 B. 32 C. 36 D. 44



PART – B

Note

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{4}$ mark.**

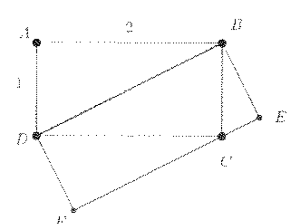
16. a and b are positive integers such that $a^2 + 2b = b^2 + 2a + 5$. The value of b is

17. After full simplification, the value of the product

$$\left(\sqrt{2+\sqrt{3}}\right)\left(\sqrt{2+\sqrt{2+\sqrt{3}}}\right)\left(\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{3}}}}\right)\left(\sqrt{2-\sqrt{2+\sqrt{2+\sqrt{3}}}}\right)$$

is ....

18. $ABCD$ is a rectangle with $AD = 1$ and $AB = 2$. $DFEB$ is also a rectangle. The area of $DFEB$ is ....



19. The two digit number whose units digit exceeds the tens digit by 2 and such that the product of the number and the sum of its digits is 144 is ....

20. If $x = \frac{p}{q}$ where p, q are integers having no common divisors other than 1, satisfies

$$\sqrt{x+\sqrt{x}} - \sqrt{x-\sqrt{x}} = \frac{3}{2}\sqrt{\frac{x}{x+\sqrt{x}}}$$

then x is ....

21. AE and BF are medians drawn to the legs of a right angled triangle ABC. The numerical value of $\frac{AE^2 + BF^2}{AB^2}$ is ....

22. AB is a chord of a circle with center O. AB is produced to C such that $BC = OA$. CO is produced to E. The value of $\frac{\angle AOE}{\angle ACE}$ is ....

23. The number of two digit numbers that are less than the sum of the squares of their digits by 11 and exceed twice the product of their digits by 5 is ....

24. AB is a diameter of circle and CD is a parallel chord. P is any point in AB. The numerical value of $\frac{PC^2 + PD^2}{PA^2 + PB^2}$ is ....

25. In the sequence $1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8, \ldots$ the 2016th term is 2^n. Then $n =$ ....

26. Each root of the equation $ax^2 + bx + c = 0$ is decreased by 1. The quadratic equation with these roots is $x^2 + 4x + 1 = 0$. The numerical value of $b + c$ is ....

27. The number of integers n such that $\frac{n+2}{n^2+1} > \frac{1}{2}$ is ....

28. P_1 and P_2 are two regular polygons. The number of sides of P_1 and P_2 respectively are in the ratio $3 : 2$ and the respective interior angles are in the ratio $10 : 9$. Then the sum of the number of sides of P_1 and P_2 is ....

29. In triangle ABC, F and E are the mid points of AB and AC respectively. P is any point on the side BC. The ratio $\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta FPE}$ is ......

30. x, y, z are distinct real numbers such that

$$x + \frac{1}{y} = y + \frac{1}{z} = z + \frac{1}{x}$$

The value of $x^2y^2z^2$ is

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